

Md Nafiu Rahman

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About Me

I am a curious and driven computer science student and researcher with a keen interest in software engineering and LLMs. I enjoy the art of teaching.

Education

Bangladesh University of Engineering & Technology (BUET), Dhaka *Feb 2020 – Mar 2025*
B.Sc. in Computer Science & Engineering CGPA: 3.81 / 4.00 (Final year: 3.94 / 4.00)
◦ Relevant coursework: Data Structures & Algorithms, Object Oriented Programming, Software Engineering, Database Systems, Numerical Methods, Machine Learning.

Notre Dame College, Dhaka *Jan 2017 – Jan 2019*
Higher Secondary Certificate GPA: 5.00 / 5.00

Bir Shrestha Noor Mohammad Public College, Dhaka *Jan 2009 – Dec 2016*
Secondary School Certificate GPA: 5.00 / 5.00

Professional Experience

Lecturer, Department of CSE, BRAC University, Dhaka *July 2025 – Present*
◦ Teaching undergraduate courses including Numerical Methods and Software Engineering. I design course materials, assignments, and practical labs with an emphasis on clear, reproducible programming and sound engineering practices.

Research Assistant, Department of CSE, BUET, Dhaka *Mar 2025 – June 2025*
Supervisor: Dr. Anindya Iqbal
◦ Worked on a pipeline to generate detailed test cases from web user interfaces using Large Language Models.

Research Experience

Empirical Study on Calibration of LLMs in Code Generation.
Authors: Mohammed Latif Siddiq, Md Nafiu Rahman, Joanna C. S. Santos.
Accepted at ICSME Research Track
Contributions: Conducted experiments across all evaluation settings and benchmarks, implemented the evaluation pipeline, and contributed to result validation across multiple models and programming languages.

Secret Breach Detection in Source Code with Large Language Models.
Authors: Md Nafiu Rahman, Sadif Ahmed, Zahin Wahab, Rifat Shahriyar, S. M. Sohan.
(ESEM 2025 Technical Track) [🔗](#)
Accepted at ESEM 2025 Technical Track
We propose a hybrid LLM-based framework for secret detection in source code, combining regex-based extraction with LLM classification to reduce false positives. Fine-tuned open-source models, including LLaMA-3.1 8B and Mistral-7B, achieve up to 0.985 F1 and 0.982 accuracy on a large GitHub benchmark.
Contributions: Developed the complete end-to-end pipeline, conducted experiments across all evaluation settings, and formulated detailed analyses.

Secret Leak Detection in Software Issue Reports using LLMs: A Comprehensive Evaluation.
Authors: Sadif Ahmed, Md Nafiu Rahman, Zahin Wahab, Rifat Shahriyar, Gias Uddin.
(MSR 2026 Technical Track) [🔗](#)
Accepted at MSR 2026 Technical Track
We introduce a hybrid pipeline for detecting secret leaks in GitHub issue reports using regex extraction and contextual LLM classification. Fine-tuned models such as Qwen and LLaMA achieve up to 94.49% F1 and generalize well to real-world repositories (81.6% F1).

Contributions: Conducted the motivational survey, curated and annotated the dataset, developed the full detection pipeline, ran experiments across all models and settings, and formulated the evaluation and analysis.

IssueGuard: Real-Time Secret Leak Prevention Tool for GitHub Issue Reports.

Authors: Md Nafiu Rahman, Sadif Ahmed, Zahin Wahab, Rifat Shahriyar, Gias Uddin.

[FSE 2026 Tool Demonstrations Track](#) [🔗](#)

Accepted at FSE 2026 Tool Demonstration Track

We introduce IssueGuard, a real-time tool that detects sensitive information in GitHub issues before posting. Leveraging a fine-tuned CodeBERT model, LRU caching, and AMP precision, the system offers fast and accurate inference. A user study with 50 participants demonstrates its effectiveness and usability.

Contributions: Developed the complete tool, designed an efficient backend with caching, asynchronous threading, and mixed precision inference, and implemented the end-to-end browser extension.

BanglaForge: LLM Collaboration with Self-Refinement for Bangla Code Generation.

Authors: Mahir Labib Dihan, Sadif Ahmed, Md Nafiu Rahman.

[\(BLP 2025\)](#) [🔗](#)

Accepted at BLP Workshop at ACL-IJCNLP 2025

We introduce BanglaForge, a framework to generate executable code from Bangla natural language descriptions using retrieval-augmented dual-model collaboration and iterative self-refinement. The system achieves 84.00% Pass@1 accuracy on the BLP-2025 benchmark.

Contributions: Contributed to developing the pipeline, created the glossary for language translation and prepared the dataset for RAG-based approach.

A Survey on Agentic Security: Applications, Threats and Defenses.

Authors: Asif Shahriar, Md Nafiu Rahman, Sadif Ahmed, Farig Sadeque, Md Rizwan Parvez.

[\(arXiv\)](#) [🔗](#)

Submitted at EACL 2026

The first comprehensive survey of agentic security, reviewing 150+ papers from 2024 and 2025. The work organizes the field into three pillars, Applications, Threats, and Defenses, providing a unified framework to understand LLM agents in cybersecurity contexts.

Contributions: Contributed to creating the taxonomy and systematically gathering and categorizing papers for the survey.

Explainable Transformer-CNN Hybrid for Modeling Brain Aging from MRI Images.

Authors: Wasif Jalal, Md Nafiu Rahman, Md Sohel Rahman.

[\(arXiv\)](#) [🔗](#)

In preparation for submission

Ongoing research on hybrid Transformer-CNN models for brain-age prediction, emphasizing explainability and interpretable fusion of slice-wise and volumetric MRI representations.

Contributions: Contributed to pipeline design, MRI preprocessing, and running experimental evaluations.

EVCC: Enhanced Vision Transformer-ConvNeXt-CoAtNet Fusion with Adaptive Routing for Classification.

Authors: Kazi Reyazul Hasan, Md Nafiu Rahman, Sadif Ahmed, Wasif Jalal, Shahriar Raj, Mubasshira Musarat, Muhammad Abdullah Adnan.

[\(Paper\)](#)

Ongoing

EVCC is a multi-branch hybrid architecture combining Transformers and convolutional backbones through adaptive token pruning, gated cross-attention, and dynamic routing. The model improves accuracy-efficiency trade-offs with significant FLOP reductions.

Contributions: Contributed to running and managing experimental evaluations.

Technical Skills

Data Science & ML: Python, NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, Torchvision.

Databases: PostgreSQL, PL/pgSQL, MongoDB, Firebase Firestore.

Full-Stack: Node.js (backend), HTML/CSS, React, Svelte, Flutter.

Languages: C/C++, Python, Java, JavaScript / TypeScript, PHP, Bash, Dart.

Language Proficiency

Bengali Native

English IELTS Overall 8.0 – Listening: 9, Reading: 9, Writing: 7, Speaking: 6.5

Achievements

- Top 20 finalists Robi Datathon 2024 (national deep learning competition).
- Deans list award and university merit scholarship at BUET.
- 5th at BLP 2025 Code Generation Challenge

Academic Projects

- **Machine Learning Algorithms and Neural Network from Scratch** – github.com/nafiurrahman00/CSE-472 – Implementations of core ML algorithms (logistic regression with ensembles, PCA/SVD for reconstruction, EM clustering) and a feed-forward neural network with Adam optimizer built from numpy.
- **Cryptography and Security Attacks** – github.com/nafiurrahman00/CSE-406 – Implemented AES encryption and Diffie-Hellman key exchange, socket communication demos, and reviewed a mobile pentesting framework as part of course project.
- **Network Simulation** – github.com/nafiurrahman00/CSE-322 – Implemented Congestion Control Algorithm, threaded server-client sockets, error correction algorithms and simulated wired/wireless mobility scenarios.
- **Operating System Internals with xv6** – github.com/nafiurrahman00/CSE-314 – Implemented threading and synchronization primitives, system calls, and explored scheduler internals in xv6.
- **BusBuddy (Android)** – github.com/nafiurrahman00/BusBuddy-Client-End – Flutter app with Node.js backend, PostgreSQL and Firebase integration. Provided ticketing, schedules, tracking and real-time updates for university bus users.
- **Nishorgo (E-commerce)** – [Term-Project-2-2-Nishorgo](https://github.com/nafiurrahman00/term-project-2-2-nishorgo) – Full stack e-commerce site for plant sales with filtering, cart, admin analytics and order management.
- **Compiler (subset of C)** – github.com/nafiurrahman00/Compiler – Subset-of-C compiler using Lex/Yacc and 8086-style assembly generation: lexer, parser, and intermediate code generation.
- **Catch the Egg (Game)** – github.com/nafiurrahman00/Catch-The-Egg – OpenGL / Igraphics game for catching falling eggs; implemented game mechanics, scoring and difficulty scaling.

References

- **Dr. Rifat Shahriyar**, Professor, Department of CSE, Bangladesh University of Engineering and Technology (BUET) – rifat.shahriyar@gmail.com – rifat@cse.buet.ac.bd
- **S. M. Sohan**, Senior Engineer, Google – sohan39@gmail.com
- **Dr. Md Rizwan Parvez**, Scientist, Qatar Computing Research Institute (QCRI) – rizwan@ucla.edu – rizwan.incipient@gmail.com
- **Dr. Farig Sadeque**, Associate Professor, Department of CSE, BRAC University – farig.sadeque@bracu.ac.bd